

# Class 5: Data Viz with ggplot2

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## Background

There are many graphics systems in R for making plots and figures. These include so-called “*base R*” *graphics* like the `plot()` function and add on packages like **ggplot2**.

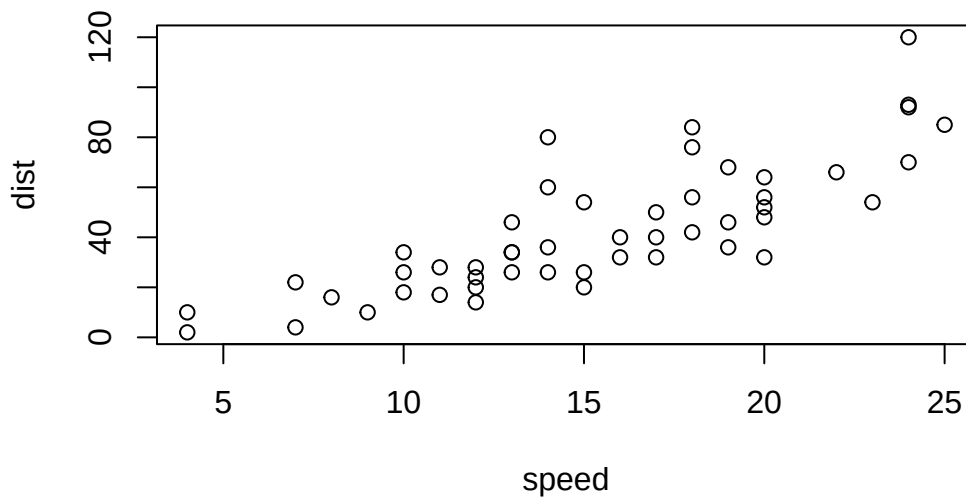
Lets compare how we make a simple figure with these two systems:

We can use the in-built `cars` dataset:

```
head(cars)
```

```
      speed dist
1         4    2
2         4   10
3         7    4
4         7   22
5         8   16
6         9   10
```

```
plot(cars)
```

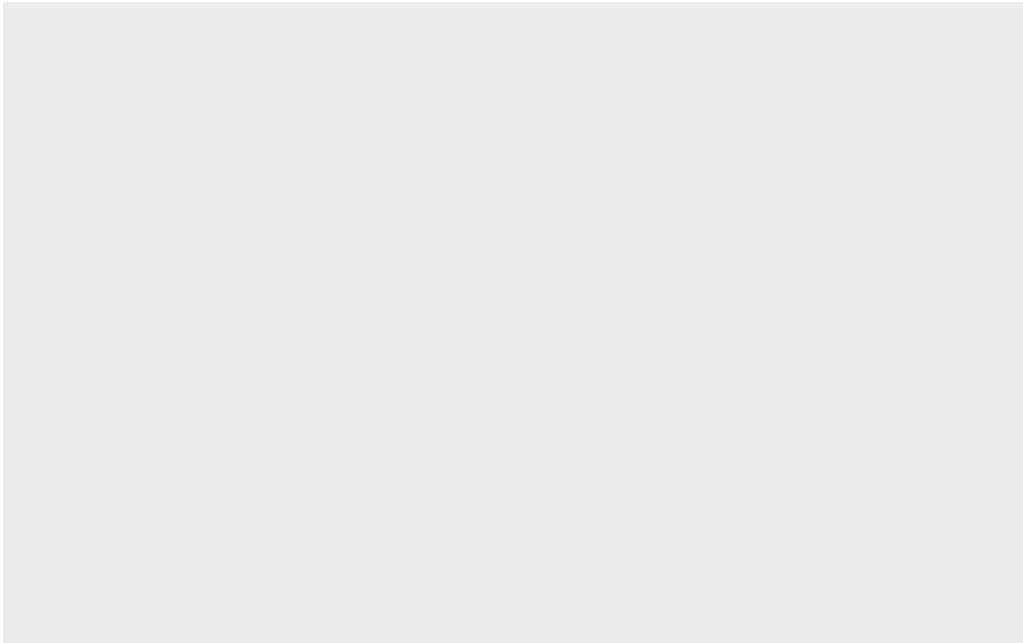


Before I can use ggplot2 I need to install it on my computer. To do this, we can use the function `install.packages("ggplot2")`. The main function in **ggplot2** to package is called `ggplot()`.

**N.B** we never run `install.packages()` into a quarto docu (run it once in the R console to prevent rerunning it over and over again) this will prevent a re-installation of the package every time the “render” is run.

Once ggplot2 is installed we need to load up the package into our R brain

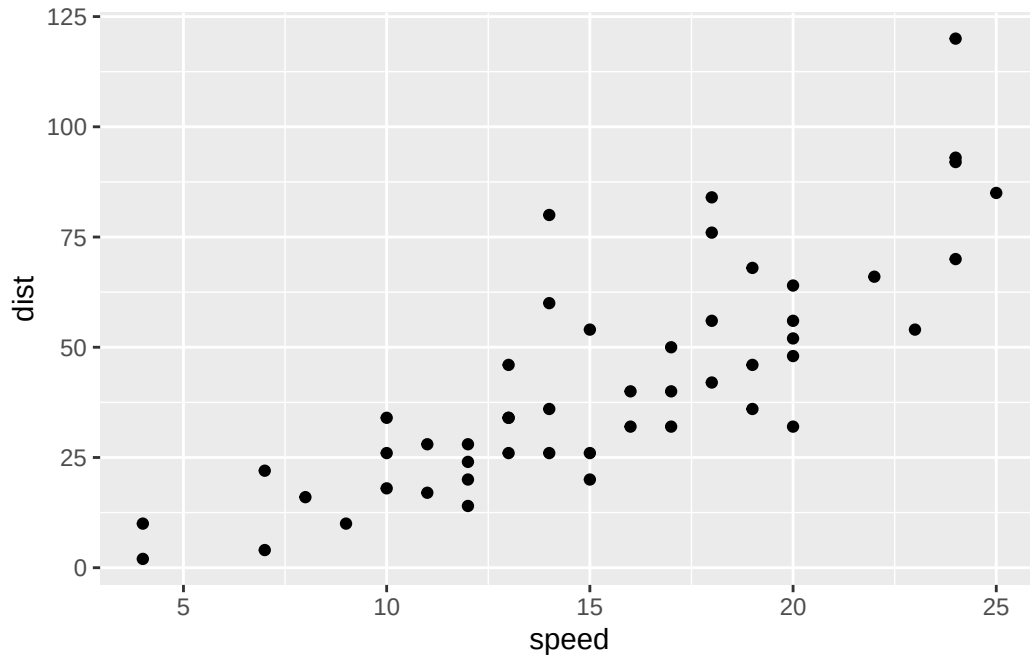
```
library(ggplot2)
ggplot(cars)
```



Every ggplot has at least 3 layers :

- the **Data** (a data.frame of the stuff we want to plot)
- the **aesthetics** (how the data maps to the plot)
- the **geometry** layer (how you want the plot drawn, e.g. points, lines, etc.)

```
ggplot(cars) +  
  aes(x= speed, y= dist) +  
  geom_point()
```

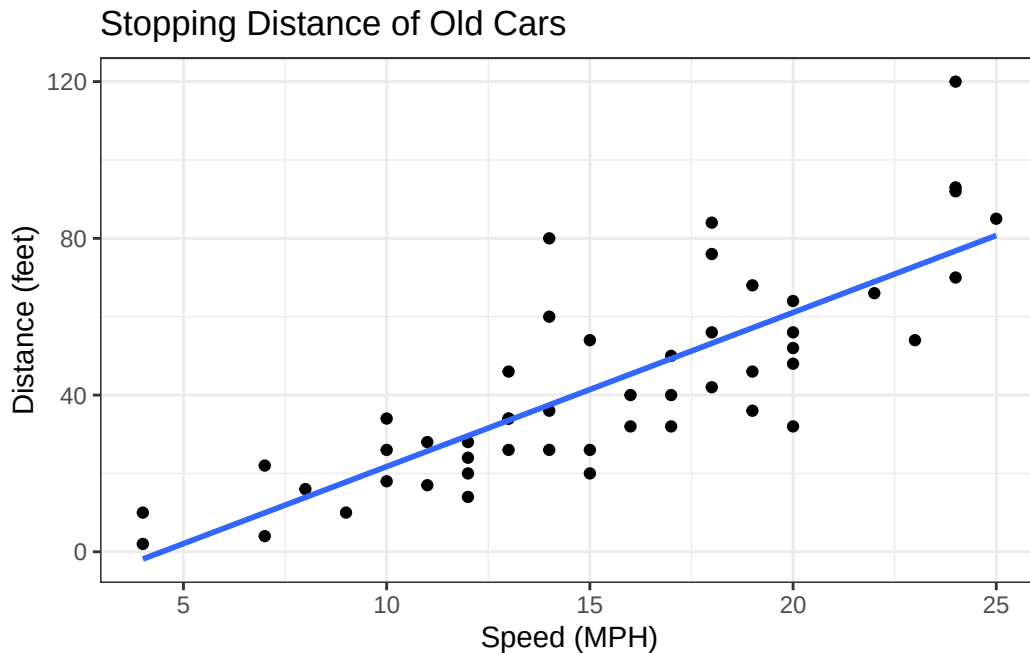


### Add some custom features to the plot

Let's add a trend line that shows the relationship between speed and distance.

```
ggplot(cars) +  
  aes(x= speed, y= dist) +  
  geom_point() +  
  geom_smooth(method = lm, se = FALSE) +  
  theme_bw() +  
  labs(title = "Stopping Distance of Old Cars",  
        x = "Speed (MPH)", y = "Distance (feet)")
```

`geom\_smooth()` using formula = 'y ~ x'



Q. Can you make the `geom_smooth()` produce a linear straight line fit to the data set and turn off the “gray” error region

## Gene expression figure

Importing the data

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

|   | Gene       | Condition1 | Condition2 | State      |
|---|------------|------------|------------|------------|
| 1 | A4GNT      | -3.6808610 | -3.4401355 | unchanging |
| 2 | AAAS       | 4.5479580  | 4.3864126  | unchanging |
| 3 | AASDH      | 3.7190695  | 3.4787276  | unchanging |
| 4 | AATF       | 5.0784720  | 5.0151916  | unchanging |
| 5 | AATK       | 0.4711421  | 0.5598642  | unchanging |
| 6 | AB015752.4 | -3.6808610 | -3.5921390 | unchanging |

```
sum(genes$State == "up")
```

```
[1] 127
```

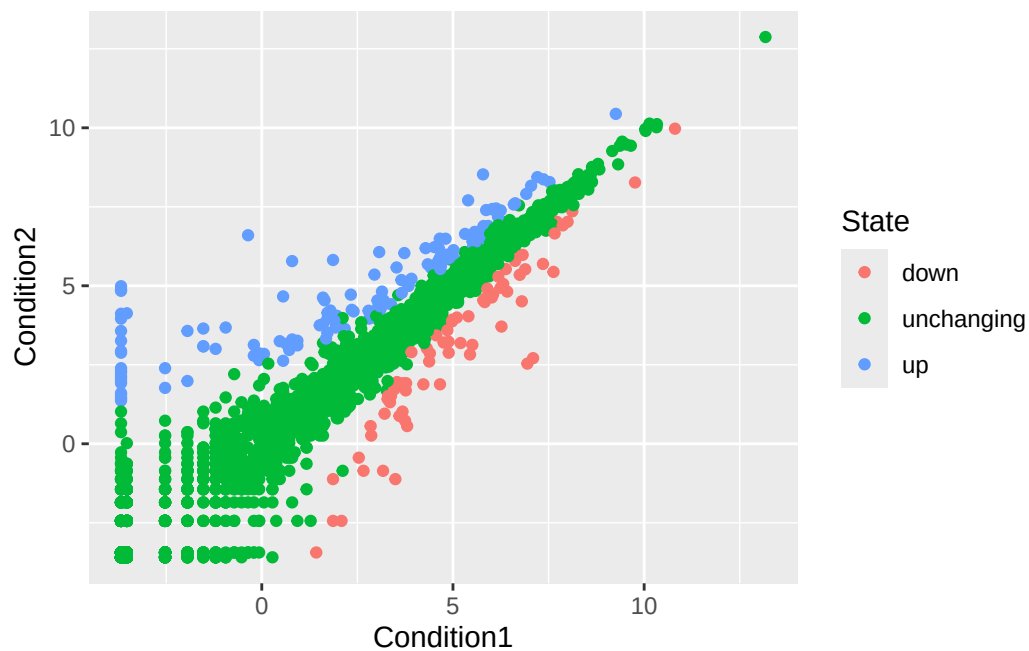
A useful new function in this context is the `table()` function:

```
table(genes$State)
```

| down | unchanging | up  |
|------|------------|-----|
| 72   | 4997       | 127 |

First Plot Attempt

```
ggplot(genes) +  
  aes(x= Condition1, y= Condition2, col= State) +  
  geom_point()
```



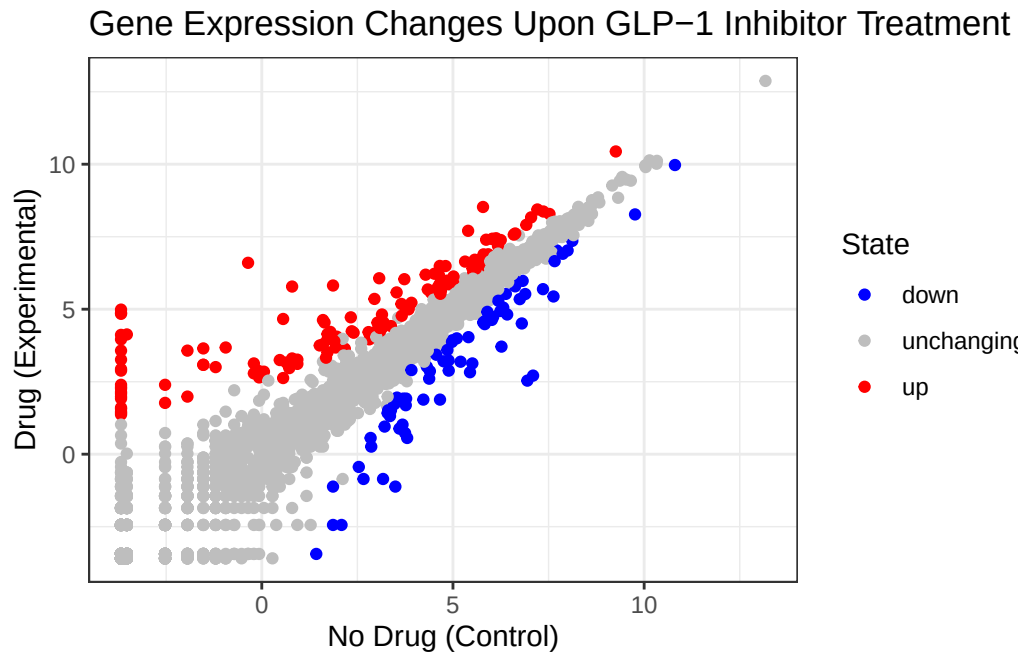
Second Plot Attempt

```
colors = c("blue", "grey", "red")  
  
ggplot(genes) +
```

```

aes(x= Condition1, y= Condition2, col= State) +
geom_point() +
scale_color_manual(values = colors) +
theme_bw() +
labs(x= "No Drug (Control)", y= "Drug (Experimental)",
      title = "Gene Expression Changes Upon GLP-1 Inhibitor Treatment")

```



### Going Further: Measures of socioeconomic factors that contribute to GLP-1 evaluations

```

# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv"

gapminder <- read.delim(url)
head(gapminder)

```

|   | country     | continent | year | lifeExp | pop      | gdpPercap |
|---|-------------|-----------|------|---------|----------|-----------|
| 1 | Afghanistan | Asia      | 1952 | 28.801  | 8425333  | 779.4453  |
| 2 | Afghanistan | Asia      | 1957 | 30.332  | 9240934  | 820.8530  |
| 3 | Afghanistan | Asia      | 1962 | 31.997  | 10267083 | 853.1007  |

|   |             |      |      |        |          |          |
|---|-------------|------|------|--------|----------|----------|
| 4 | Afghanistan | Asia | 1967 | 34.020 | 11537966 | 836.1971 |
| 5 | Afghanistan | Asia | 1972 | 36.088 | 13079460 | 739.9811 |
| 6 | Afghanistan | Asia | 1977 | 38.438 | 14880372 | 786.1134 |

Q. How many entries (i.e. rows) are in this dataset?

```
nrow(gapminder)
```

```
[1] 1704
```

q. How many different “countries” are entered in the dataset?

```
length( table(gapminder$country) )
```

```
[1] 142
```

```
length( unique(gapminder$country) )
```

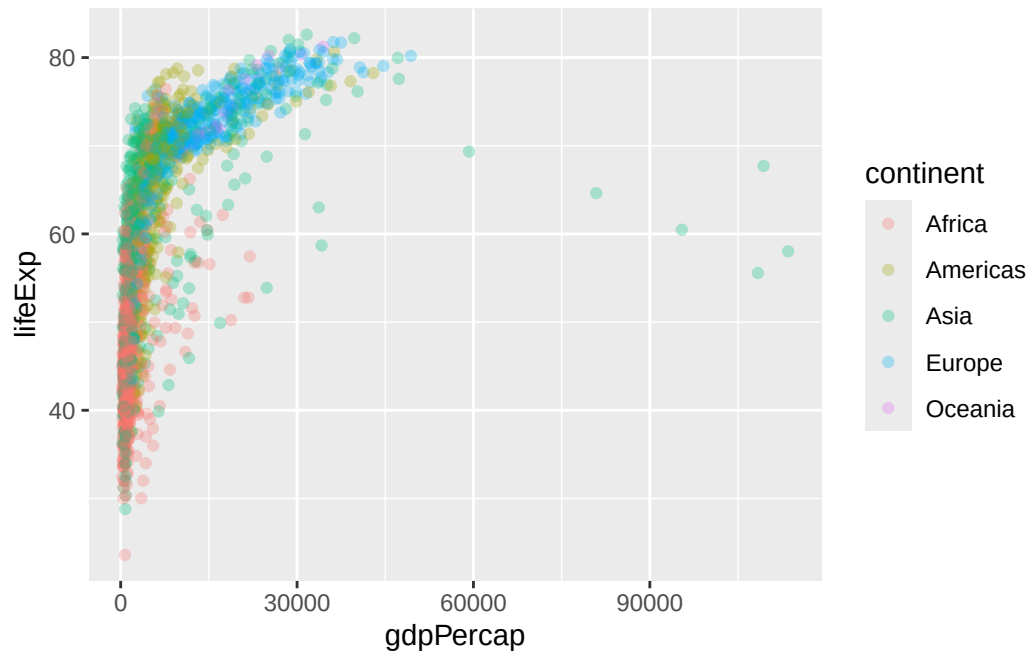
```
[1] 142
```

First Plot Being Created of the Entire Dataset for GDP Per Capita vs. Life Expectancy

```
p <- ggplot(gapminder) +
  aes(x = gdpPercap, y = lifeExp, col = continent) +
  geom_point(alpha = 0.3)
```

```
p
```





Making a plot for 1977 and 2007 ONLY

Q. First use the **dplyr** package and the **filter()** function from that package to extract the rows from the year 2007.

```
library(dplyr)
```

```
g07 <- filter(gapminder, year == 2007)
g77 <- filter(gapminder, year == 1977)
g <- filter(gapminder, year == 2007 | year == 1977)
```

```
p <- ggplot(g) +
  aes(x = gdpPerCap, y = lifeExp, col = continent, size = pop) +
  geom_point() +
  facet_wrap(~year)
```

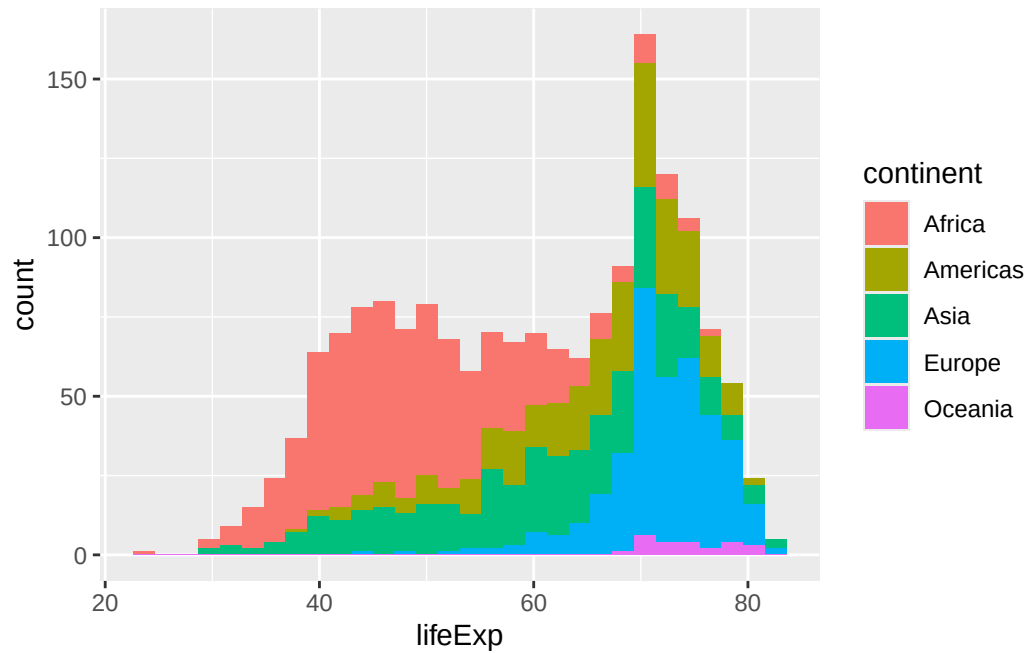
p



Q. Make a histogram of lifeExp colored by continent Q. Make a histogram of lifeExp faceted by continent

```
ggplot(gapminder) +
  aes(lifeExp, fill = continent) +
  geom_histogram()
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



```
ggplot(gapminder) +  
  aes(lifeExp, col = continent) +  
  geom_histogram() +  
  facet_wrap(~continent)
```

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.

